RNA Bioinformatics M2 Projects — Student Briefs

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Project briefs for the Master 2 GENIOMHE-AI course in RNA bioinformatics, academic year 2026–2027. This document specifies sixteen semester-long research projects. Each brief states a research question, the data and tools, the evaluation metrics, a sixteen-week work plan, a risk analysis, a scope table and a novelty-and-reproducibility block. Projects are sized for part-time teams and designed so that the scaled-down version delivered in week 8 is a standalone result.

Scientific ambition. These projects are designed as genuine research projects, not training exercises. The objective is a scientifically rigorous and potentially novel result that, if the work succeeds, could contribute to the literature and form the basis of a manuscript submitted to a peer-reviewed journal. To keep that step short, every final report is written in the form of a research article.

Publication is not guaranteed: it depends on the novelty, robustness and significance of the results, and on the quality of the execution. Nor is it a requirement for passing the course. A well-conducted project that yields a meaningful negative result or a useful research resource is a successful project.

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How to use these briefs

Each brief is sized for one semester: 16 weeks, part-time, with no upper limit on team size (the overview gives recommended sizes, not eligibility limits). Every brief sorts its work into three classes: the required core, delivered as the week-8 minimum viable version (MVP); one main extension for weeks 9–13; and stretch work attempted only after the main extension is complete. Difficulty is rated 2 (low: mature tools, clear evaluation), 3 (moderate) or 4 (high: several hard components, scope must be cut early). No brief depends on AlphaFold 3 weights; where a tool is optional, the brief says so. Numbering follows the original project list (8 and 13 were removed). Claims about tool inputs and training cutoffs were checked on 16–17 October 2026 against the documentation linked in the References section; teams re-check them in week 1.

Shared milestones (every brief uses the same skeleton)

- Week 1: roles assigned, environment installed, and an end-to-end smoke test on three representative inputs. For GPU briefs the test must show inference actually running, with runtime, GPU memory and storage per run recorded. A tool whose documented inputs cannot support the planned experiment is replaced now, not rescued in week 8.
- Week 2: literature refresh for the exact question, datasets and metrics (closest prior work, including 2025–2026 papers), written into the novelty block, which seeds the report's introduction; go/no-go on any tool or dataset still unverified.
- Week 3: protocol frozen and written down: tool versions, dataset releases, inclusion rules, metrics, hypotheses, statistical tests. Changes after week 3 are logged with a reason. The protocol is the first draft of the report's methods section.
- Week 8: MVP reviewed by the instructor; a team that is behind cuts scope here, not later.
- Week 14: results frozen; weeks 15–16 go to the article-form report, the repository and the presentation.

Roles (scale with the team)

- Harness owner: evaluation code, tests on known answers, result tables.
- Data owner: curation, provenance, release versions, exclusion log.
- Tool runners: one per predictor or pipeline, responsible for versions and run logs.
- Writer: protocol document, weekly log, report.

In a team of two or three the roles rotate. A team larger than four splits independent work packages (one predictor, one data source, one analysis) among subgroups, under one shared protocol, one evaluation harness and one integrated report. More students mean more parallel runs and more validation, not a wider scientific scope.

Deliverables for every project

- Versioned repository with environment files (conda or Docker), run scripts and result tables.
- Protocol document (frozen in week 3) and its change log.
- The scope table (required, main extension, stretch) and the completed novelty-and-reproducibility block, both frozen with the protocol.

- Final report written as a research article: abstract, introduction, methods, results, discussion and references, with figure legends, a data and code availability statement, and a supplementary table of every tool version and dataset release used.
- A 15-minute presentation.

Scope classes (every brief uses them)

- Required: the minimum analysis that answers the main question; it is the MVP.
- Main extension: one planned addition that strengthens the contribution.
- Stretch: further tools, data types or sensitivity analyses, attempted only after the main extension is complete. Stretch work is never an implicit requirement for a successful project.

Novelty and reproducibility block (every brief carries one, pre-filled; teams complete it by week 3)

Field	What the team establishes
Closest prior work	Three to five relevant papers, including the most recent related work
Specific new contribution	What the project measures, compares or releases that prior work did not
Main endpoint	The single primary quantitative result that answers the central question
Evaluation risks	Leakage, mismatched constructs, missing labels, training-data overlap, other confounders
Feasibility evidence	Confirmed data access, a working pilot, supported input formats, a realistic compute estimate

Overview

Brief	Difficulty	Compute	Recommended team
1 Temporal test of co-folding models	4	GPU, heavy	4–5
2 Docking vs co-folding, RNA–ligand	4	GPU, moderate	4–5
3 Riboswitch switching	3	GPU, light	3–4
4 Modified nucleotides as input	2	GPU, light	2–3
5 Stereochemistry and refinement	2	CPU	2–3
6 Model selection	3	CPU (GPU optional)	3
7 Calibration of pair probabilities	2	CPU	2–3
9 MSA ablation	3	CPU heavy, then GPU	3–4
10 Leakage audit of LM benchmarks	2	CPU (GPU optional)	3–4
11 Replication study	3	Depends on the paper	2–4
12 Systematic review	2	None	4–6
14 Modification evidence atlas	3	None	3–4
15 Rfam structural dark matter	3	GPU, moderate	3–4
16 Independent evaluation of inverse folding	4	GPU, heavy	4–5
17 Probing LM embeddings	3	GPU, light	3
18 Viral RNA elements	3	GPU, light	3–4

Brief 1 — Temporal, homology-stratified test of 2026 co-folding models

Difficulty 4 · Recommended team 4–5 · GPU heavy · MVP (week 8): Boltz-2 and Chai-1, run single-sequence for RNA, on post-cutoff RNA monomers, with results reported against sequence similarity and against structural similarity to pre-cutoff structures.

Question. How accurate are 2026 co-folding models on RNA structures released after their training cutoffs, and how does accuracy vary with sequence and structural similarity to the structures they were trained on? Hypothesis: accuracy falls as similarity to pre-cutoff structures falls. A decline is reported as performance versus similarity; it is not by itself proof of memorization, because undocumented training data, templates and other inputs can produce the same pattern.

Data.

- Targets: PDB entries containing RNA chains released after the latest documented training cutoff among the models compared (RCSB Search API: polymer type RNA, resolution ≤ 4 Å, chain length 20–500 nt, no nucleotide-only ligands). Cutoffs differ by model: Boltz-2's report describes training structures released before 1 June 2023; Chai-1's technical report gives January 2021. The primary set uses the latest cutoff among the models compared.
- Date semantics: keep both the initial deposition date and the first public release date for every entry. The primary test set excludes anything deposited before the cutoff; a sensitivity analysis applies the release-date rule, because whether a cutoff was applied to deposition or release dates is not always documented.
- Reference set: all RNA chains released before the cutoff, for similarity stratification.
- Clustering protocol adapted from RNA3DB (sequence- and structure-based); Rfam for family labels.
- Optional: CASP17 RNA targets if their experimental structures are released during the semester.

Tools.

- Predictors: Boltz-2 and Chai-1 (pip-installable, GPU); Protenix and OpenFold3 as the main extension; RNAPro only if alignments and templates are already in hand; AlphaFold 3 only if weights are already available.
- Alignments: only through each predictor's documented route, recorded per run. Boltz-2's input schema accepts a custom alignment for protein chains only, so its RNA chains run single-sequence; Chai-1 documents custom alignments without an RNA-specific route; Protenix (v1.0.0 and later) and AlphaFold 3 document an RNA alignment search. No alignment is supplied to a tool that does not document consuming it.
- Evaluation: US-align (TM-score), RNAdvisor (RMSD, lDDT, INF, clash), MMseqs2 and Infernal for sequence and family similarity, US-align for structural similarity to the reference set.

Metrics.

- Primary: TM-score per target; secondary: RMSD, lDDT, INF; fraction of targets with TM-score > 0.45.

- Performance versus similarity: two separate curves, TM-score against maximum sequence identity and TM-score against maximum TM-score to any pre-cutoff structure, in bins with bootstrapped confidence intervals; reported as association, not as memorization.
- Target-level paired comparisons between tools on identical targets (paired Wilcoxon test, bootstrap of the paired difference); breakdown by length, Rfam family type and presence of pseudoknots.

Week-by-week plan (item number = week).

1. Roles; GPU smoke test on three RNAs with Boltz-2 and Chai-1, recording runtime, GPU memory and storage.
2. Record each model's documented training cutoff and its date semantics (deposition or release); refresh the closest prior work; draft the PDB query.
3. Freeze the protocol: inclusion filters, cutoff rules, metrics, similarity bins, statistical tests, tool versions.
4. Build the harness (predict → align → score → table); test on five targets with known answers.
5. Build the pre-cutoff reference set; compute sequence identity and structural similarity for every target.
6. Run Boltz-2 on all monomer targets (3 seeds); store models and confidence values.
7. Run Chai-1 on all monomer targets.
8. MVP (required): results for two tools on monomers against the two similarity axes; internal review.
9. Main extension: Protenix with its documented RNA alignment search on and off, and OpenFold3 if installed; re-run the harness.
10. Stretch: RNA–RNA and protein–RNA complexes (subset) with interface metrics.
11. Analysis: similarity curves, failure modes (long RNAs, pseudoknots, multimers).
12. Case studies: three to five instructive targets with visual comparison.
13. Robustness: seeds, deposition-versus-release cutoff rule, resolution filter sensitivity.
14. Freeze results; make figures.
15. Draft the report; clean the code; write environment files.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
GPU time runs short	Restrict to monomers ≤ 200 nt and one seed per tool.
A model's cutoff or its date semantics is undocumented	Deposition-date rule as the primary analysis, release-date rule as a sensitivity analysis; both reported.
Too few post-cutoff structures	Use the oldest cutoff among evaluated models; add CASP17 targets if released.
A predictor's alignment route is unclear	Run it single-sequence and say so; never feed an alignment a tool does not document consuming.
AlphaFold 3 weights not obtained	Exclude it and say so; the study stands on open-weight models.

Scope.

Class	Work
Required (MVP)	Two open-weight predictors, single-sequence RNA, post-cutoff monomers, two similarity axes, paired statistics.
Main extension	Protenix with its documented RNA alignment search on and off; OpenFold3.
Stretch	Complexes with interface metrics; CASP17 targets; case studies.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	AlphaFold 3 for RNA benchmark (Bernard et al., 2025); the 2026 ten-method benchmark on monomers and complexes; FoldBench; RNA3DB (Szikszai et al., 2024).
Specific new contribution	2026 models on a post-cutoff set with documented date semantics, scored against both sequence and structural similarity to the pre-cutoff PDB.
Main endpoint	Median TM-score per tool in each similarity bin, with intervals.
Evaluation risks	Cutoff ambiguity, undocumented training data or templates, small bins, alignment routes that differ by tool.
Feasibility evidence	Week-1 smoke test with runtimes; the number of post-cutoff targets counted in week 2.

Brief 2 — Docking vs co-folding for RNA–small-molecule complexes

Difficulty 4 · Recommended team 4–5 · GPU moderate · MVP (week 8): co-folding-only evaluation (Boltz-2, Chai-1) on RNA–ligand complexes solved after 2024, stratified by ligand and pocket novelty, after a ten-complex pilot of the whole pipeline.

Question. On RNA–small-molecule complexes solved after 2024, do co-folding models given sequence plus SMILES place the ligand more accurately than classical docking into experimental or predicted RNA structures, and does accuracy collapse for ligands and pockets unlike anything in the training data?

Data.

- Starting set: HARIBOSS (curated RNA–small-molecule complexes), extended with RCSB entries released after 2024 (ligand mass 150–1000 Da; ions, solvents, buffers and nucleotides excluded).
- Per complex: ligand SMILES from the Chemical Component Dictionary, binding-site residues (within 5 Å), RNA chain, resolution, method.
- Novelty measures: Tanimoto similarity (Morgan fingerprints) to pre-cutoff RNA-bound ligands; pocket similarity to pre-cutoff pockets by US-align on binding-site residues.

Tools.

- Co-folding: Boltz-2 and Chai-1 with ligand SMILES; AlphaFold 3 only if weights are available (the server does not take arbitrary ligands).
- Docking: rDock (RNA-capable) and AutoDock Vina for poses; AnnapuRNA for RNA-specific rescoring; RLDOCK optional.
- Chemoinformatics: RDKit (ligand preparation, fingerprints, symmetry-aware RMSD), Open Babel.
- Evaluation: own scripts for ligand RMSD after pocket superposition; PoseBusters-style validity checks (clashes, bond geometry) adapted to RNA; US-align and RNAdvisor for the RNA fold.

Metrics.

- Pose success: symmetry-aware ligand heavy-atom RMSD ≤ 2 Å after superposition on pocket residues; centroid distance; pocket recovery (fraction of contact residues within 4 Å).
- Physical validity: clash and geometry failure rates per method.
- Three conditions, never pooled: co-folding (RNA and ligand predicted together), docking into the experimental RNA, docking into the predicted RNA. RNA-fold accuracy (TM-score) is reported beside pose accuracy so that fold errors are not confused with pose errors.
- All of the above stratified by ligand Tanimoto bin and pocket-similarity bin; bootstrapped intervals.

Week-by-week plan (item number = week).

1. Roles; smoke test: Boltz-2 in ligand mode and RDKit on one known complex, runtime and memory recorded.

2. Assemble the complex set from HARIBOSS plus post-2024 RCSB entries; apply ligand filters; refresh the closest prior work.
3. Freeze the protocol: success criteria, novelty bins, exclusion rules, the three conditions, tool versions.
4. Pilot on ten complexes: ligand preparation, stereochemistry checks, symmetry-aware RMSD, pocket mapping; fix the pipeline before scaling.
5. Curate 60–120 complexes: SMILES, binding-site residues, handling of multiple ligand copies; harness validated on the pilot set.
6. Run Boltz-2 co-folding on all complexes (3 seeds).
7. Run Chai-1 co-folding; compute novelty measures for every complex.
8. MVP (required): co-folding results stratified by novelty; internal review.
9. Main extension: rDock with RNA parameters, docking into experimental RNA structures.
10. Main extension continued: docking into co-folding-predicted RNA structures; AnnapuRNA rescoring.
11. Analysis: docking versus co-folding by condition; success versus novelty; validity failures.
12. Stretch: case studies (riboswitches, viral elements, synthetic aptamers).
13. Robustness: seeds, pocket radius, alternative RMSD definitions.
14. Freeze results; make figures.
15. Draft the report; clean the code; environment files.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
Docking tools fail to set up	Co-folding-only study on new complexes, which stands on its own.
Too few post-2024 complexes	Extend to post-2022 entries and lean on the novelty stratification.
Ligand ambiguity (multiple ligands, covalent, metal-bridged)	Pre-registered exclusion rules, logged per entry.
Large RNAs (ribosomal) overwhelm co-folding	Cap RNA length at 200 nt; report excluded entries.

Scope.

Class	Work
Required (MVP)	Co-folding on a curated post-2024 set, after the ten-complex pilot, stratified by ligand and pocket novelty.
Main extension	Docking into experimental and into predicted receptors as two separate conditions, with rescoring.
Stretch	AutoDock Vina or RLDOCK as further docking engines; case studies.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	Ribo-LENS and SMARTPocket (2026) on RNA binding-site prediction; the Runs N' Poses memorization study and the docking-versus-co-folding comparison for proteins (Röhrig et al., 2026); HARIBOSS; AnnapuRNA.
Specific new contribution	First systematic docking-versus-co-folding comparison on RNA–

Field	Starting point
	ligand complexes solved after 2024, stratified by ligand and pocket novelty.
Main endpoint	Success rate (ligand RMSD ≤ 2 Å) per method and condition, by novelty bin.
Evaluation risks	Ligand ambiguity, receptor-quality confounding, co-folding training overlap with HARIBOSS entries, small bins.
Feasibility evidence	Ten-complex pilot by week 4; the number of post-2024 complexes counted in week 2.

Brief 3 — Riboswitch conformational switching in co-folding models

Difficulty 3 · Team 3–4 · GPU light · MVP (week 8): ten riboswitch classes with apo/holo structure pairs, one predictor, run with and without the ligand declared.

Question. Do co-folding models reproduce the ligand-dependent conformational states of riboswitch aptamer domains, or do they return one state whatever the input? Secondary question: does declaring the ligand move the prediction toward the ligand-bound state, and does the model's confidence track which state it produced? A ligand-conditioned prediction is evidence about which state the model outputs, not about the physical switching process or the equilibrium between states; the brief measures the former.

Data.

- Riboswitch classes with both ligand-free and ligand-bound experimental structures in the PDB (candidates: purine, SAM-I/II/III, TPP, lysine, glycine, c-di-GMP, FMN, preQ1, glmS, ZTP, fluoride, guanidine, THF, Mn2+), collected by Rfam family; NMR or cryo-EM apo structures where no apo crystal exists.
- Inclusion rules, fixed in week 3: the apo and holo structures cover the same aptamer domain with matched construct boundaries; a residue-level sequence mapping between the two constructs; apo structures bound to ligand analogues, or whose crystal contacts enforce the holo fold, are excluded or flagged.
- A pair table per class: Rfam ID, apo PDB, holo PDB, ligand, resolution, method, construct boundaries, residue mapping.
- Ligand SMILES from the Chemical Component Dictionary.

Tools.

- Predictors: Boltz-2 and Chai-1 (ligand co-folding); AlphaFold 3 server for ligand-free runs only, if used.
- Structure comparison: US-align, RNAdvisor; base-pair annotation with RNApdbee or barnaba; ViennaRNA for 2D baselines.

Metrics.

- ΔTM = TM(prediction, holo) − TM(prediction, apo), computed on the same aligned aptamer domain, with and without the ligand declared; the same with pocket-local RMSD.
- State-specific base pairs: recovery of pairs present only in the holo structure and only in the apo structure.
- Ligand effect separated from template effect: every condition is run with templates disabled and enabled, so a holo-like output is attributed to the ligand only when it appears without templates.
- Ligand pose RMSD when the ligand is declared; confidence (pLDDT, ipTM) against the state produced; seed-to-seed diversity.

Week-by-week plan (item number = week).

1. Roles; literature list of classes with apo/holo structures; smoke test with Boltz-2 on one riboswitch, runtime recorded.
2. Build the pair table with construct boundaries and residue mapping; collect ligand SMILES; refresh the closest prior work.

3. Freeze the protocol: inclusion rules, comparisons, metrics, template settings, hypotheses, tool versions.
4. Prepare inputs; harness for ΔTM, pocket RMSD and state-specific pair recovery on the aligned domain.
5. Run Boltz-2 on the first ten classes: without and with ligand, templates disabled (5 seeds each).
6. Repeat with templates enabled; run Chai-1 in the same four conditions.
7. Extract 2D structures from predictions and references; define state-specific pairs per class.
8. MVP (required): ΔTM and pair-recovery table for ten classes, two tools, four conditions; internal review.
9. Main extension: all remaining classes.
10. Analysis: ligand effect versus template effect; confidence versus the state produced; seed diversity.
11. Case studies: classes with large apo/holo differences (purine, lysine, c-di-GMP).
12. Stretch: full-length constructs with expression platform; 2D ON/OFF predictions as a baseline.
13. Robustness: seeds, alternative state definitions, domain-boundary choices.
14. Freeze results; make figures.
15. Draft the report; clean the code.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
Apo structures are rare or partial	Define apo-like states from NMR ensembles or flagged analogue-bound structures; report the definition and a sensitivity analysis without flagged pairs.
The model copies a holo template from the PDB	Templates disabled is the primary condition; templates enabled is reported beside it.
Construct boundaries differ between apo and holo	Compare only on the mapped, shared aptamer domain; exclude pairs without a clean mapping.
The notion of state is contested	Pre-registered base-pair definition of state; sensitivity analysis on the pair set.
GPU time is short	Three seeds instead of five; fewer classes.

Scope.

Class	Work
Required (MVP)	Ten classes, two tools, four conditions (ligand on/off × templates on/off), on matched aptamer domains.
Main extension	All remaining classes with apo/holo pairs.
Stretch	Full-length switching with expression platforms; AlphaFold 3 server ligand-free runs.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	CASP16 structure-provider assessment (functionally important regions poorly predicted); CASP17 ensemble targets; RNA-Puzzles Round V riboswitch targets; riboswitch examples in the co-folding model papers.

Field	Starting point
Specific new contribution	A systematic apo/holo benchmark with matched constructs and a template-controlled test of ligand conditioning.
Main endpoint	Fraction of classes where ΔTM changes sign when the ligand is declared, templates disabled.
Evaluation risks	Apo-state scarcity, construct mismatch, template leakage, crystal-packing artefacts.
Feasibility evidence	Pair table of at least ten classes with residue mappings by week 2.

Brief 4 — Modified nucleotides as input to structure predictors

Difficulty 2 · Team 2–3 · GPU light · MVP (week 8): paired comparison, modified versus unmodified input, on at least 20 isolated tRNA structures with one predictor.

Question. Does declaring an RNA's post-transcriptional modifications (as modified residues in the input) improve the predicted 3D structure compared with the unmodified sequence, globally and around the modified sites?

Data.

- PDB structures at ≤ 3 Å whose RNA chains carry at least three modified residues: isolated tRNAs first, then tRNAs in complexes (synthetases, elongation factors), snRNA fragments and rRNA helices; found by querying for non-standard nucleotide components.
- MODOMICS modification maps as a cross-check of positions and types; GtRNAdb for tRNA identities.
- Chemical Component Dictionary codes for each modification (for example PSU, 5MU, 1MA, 2MG, M2G, 7MG, H2U, OMC, OMG).

Tools.

- Predictors: Boltz-2 (modified residues declared by CCD code through its modifications field, documented for protein, DNA and RNA chains); Chai-1 and AlphaFold 3 as extensions, after checking which codes each accepts.
- Week-1 check: the list of CCD codes each tool accepts, and inspection of the output coordinates to confirm the modified residue is actually built; unsupported codes are excluded, never substituted.
- Mapping script: PDB or MODOMICS positions → input file with CCD codes, validated by sequence alignment.
- Evaluation: US-align, RNAdvisor (global metrics), own script for local lDDT within ±3 nt of each modified site.

Metrics.

- Paired difference (modified − unmodified) in TM-score, RMSD, lDDT and INF per structure; paired Wilcoxon test with effect size.
- Local lDDT around modified sites; accuracy of the tRNA elbow and anticodon loop.
- Confidence changes; stratification by number and type of modifications; whether a tool silently ignores a code (checked in the output coordinates).

Week-by-week plan (item number = week).

1. Roles; smoke test: one tRNA predicted with and without modifications in Boltz-2; the list of supported modification codes verified and the output inspected.
2. Query the PDB for RNA entries with modified residues; cross-check against MODOMICS; build the structure list; refresh the closest prior work.
3. Freeze the protocol: hypotheses, metrics, paired tests, exclusions, supported-code list, tool versions.
4. Mapping pipeline from positions to CCD-coded inputs; validate on five structures.
5. Harness for global and local metrics; test on five structures.
6. Run Boltz-2 on unmodified inputs (5 seeds) for all targets.

7. Run Boltz-2 on modified inputs for all targets.
8. MVP (required): paired comparison on isolated tRNAs; internal review.
9. Main extension: a second predictor (Chai-1 or AlphaFold 3) and snRNA and rRNA fragments.
10. Analysis: global versus local effects; which modification types matter (m1A9, dihydrouridine, Ψ55, wybutosine).
11. Stretch: tRNA–protein complexes; ablations declaring only loop or only core modifications.
12. Case studies with visualizations; verify again that modifications appear in the output coordinates.
13. Robustness: seeds, templates on/off, MSA on/off where the tool documents it.
14. Freeze results; make figures.
15. Draft the report; clean the code.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
A tool does not accept a given code	Restrict the study to supported codes or exclude the case; a chemically different modification is never substituted.
A tool accepts the code but ignores it	Caught by inspecting output coordinates in week 1; such tools are reported as non-consuming and dropped from the paired test.
MODOMICS and PDB positions disagree	Treat the residues present in the deposited structure as ground truth.
No measurable effect	A null result with confidence intervals is the deliverable; report it as such.
Few isolated tRNA structures	Add tRNA chains extracted from complexes, flagged by context.

Scope.

Class	Work
Required (MVP)	Isolated tRNAs, Boltz-2, paired modified-versus-unmodified comparison with target-level tests.
Main extension	A second predictor; snRNA and rRNA fragments.
Stretch	tRNA–protein complexes; subset ablations by modification type.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	The co-folding model papers (modified residues supported as input, not evaluated for RNA); MODOMICS; CASP16 assessments, which ignore modifications.
Specific new contribution	First paired quantification of the effect of declaring modifications on RNA 3D accuracy, globally and at modified sites.
Main endpoint	Paired median ΔlDDT (modified − unmodified) at modified sites, with interval.
Evaluation risks	Codes silently ignored, position-mapping errors, isolated versus complexed context.
Feasibility evidence	Supported-code list and at least 20 isolated tRNA structures

Field	Starting point
	confirmed by week 2.

Brief 5 — Stereochemistry and refinement of deep-learning RNA models

Difficulty 2 · Recommended team 2–3 · CPU · MVP (week 8): stereochemistry statistics across public model pools plus a before/after comparison with one refinement tool on 150 models, geometry and accuracy reported as separate axes.

Question. How often do current deep-learning RNA models violate stereochemical standards (clashes, bond and angle outliers, sugar puckers, backbone suite conformers), and does cheap refinement fix the geometry without degrading, or while improving, accuracy against the experimental structure?

Data.

- Public pools: CASP16 RNA server models and RNA-Puzzles Round V submissions, each with the experimental reference.
- Own pool: Boltz-2 and Chai-1 predictions on a benchmark set (shared with Brief 1 if both run).
- Metadata per model: method family (deep learning, fragment assembly, physics, human-guided), target length, reference resolution.

Tools.

- Validation: MolProbity (clashscore, bond and angle outliers, suite conformers), RNAdvisor (clash score, lDDT, INF, RMSD), barnaba for backbone torsions and sugar puckers.
- Refinement: QRNAS (fast, knowledge-based); SimRNA restrained refinement or a short OpenMM minimization with an AMBER RNA force field as a second protocol.

Metrics.

- Geometry: clashscore, percentage of bond, angle, suite and pucker outliers, by method family and target length.
- Refinement effect: Δclashscore, ΔRMSD, ΔlDDT, ΔINF relative to the reference, computed on the same mapped residues before and after; fraction of models that improve versus degrade on each axis separately; runtime per model.
- Geometry and accuracy stay separate axes: a cleaner model is not called more native-like unless the accuracy metrics move too.
- Correlation between geometry quality and accuracy before refinement.

Week-by-week plan (item number = week).

1. Roles; smoke test: MolProbity (or its server workflow), RNAdvisor and barnaba on three models.
2. Collect model pools and metadata; organize storage and naming; refresh the closest prior work.
3. Freeze the protocol: metrics, residue mapping, the one required refinement protocol, hypotheses, comparisons.
4. Validation harness over all models; first geometry table.
5. Analysis 1: stereochemistry by method family and length.
6. Install QRNAS; refine 50 models; measure deltas on mapped residues.
7. Extend QRNAS refinement to 150 models; check the residue mapping on every pair.

8. MVP (required): geometry statistics across pools plus QRNAS before/after on 150 models; internal review.
9. Main extension: a second protocol (SimRNA restrained refinement or OpenMM minimization) on the same 150 models.
10. Analysis 2: does refinement help accuracy; which error classes are fixed.
11. Geometry-versus-accuracy correlation; is a clean model more often an accurate one?
12. Stretch: full-pool refinement; case studies of worst offenders and of refinement successes and failures.
13. Robustness: refinement parameters, two random seeds, alternative clash definitions.
14. Freeze results; make figures.
15. Draft the report; clean the code.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
MolProbity will not install	Use its server in batches, plus RNAdvisor clash score and barnaba torsions.
SimRNA refinement too slow	QRNAS only, or OpenMM minimization on a subset.
Public pools not downloadable in bulk	Own predictions only, on a larger target set.
Refinement degrades accuracy	That is a finding; report it with the geometry gains.

Scope.

Class	Work
Required (MVP)	Geometry statistics across the pools plus one refinement tool on 150 models, both axes on mapped residues.
Main extension	A second refinement protocol on the same 150 models.
Stretch	Full-pool refinement; case studies.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	Stereochemical analysis of 1,052 RNA-Puzzles models (Szachniuk group); RNAdvisor 2 (November 2025) metrics and meta-metrics; MolProbity RNA validation.
Specific new contribution	First geometry audit of 2026 co-folding outputs, and a controlled refinement test that keeps geometry and accuracy on separate axes.
Main endpoint	Fraction of models whose lDDT improves after refinement, reported beside the fraction whose clashscore improves.
Evaluation risks	Residue-mapping errors between model and reference, pool availability, refinement parameters.
Feasibility evidence	CASP16 server models downloaded by week 2; MolProbity running in week 1.

Brief 6 — Model selection in the co-folding era

Difficulty 3 · Team 3 · CPU (GPU optional) · MVP (week 8): on public CASP16 pools, predictor confidence versus two dedicated quality assessors versus a consensus score, with contamination documented.

Question. When several predictors each return several models for the same RNA, which signal best picks the most accurate one: the predictors' own confidence (pLDDT, ipTM, ranking score), a dedicated RNA model-quality assessor, or agreement among the models?

Data.

- CASP16 RNA server model pools with target structures; RNA-Puzzles Round V model sets with references.
- Own multi-predictor pools: Boltz-2, Chai-1 and Protenix on 30–50 shared targets, five seeds each.
- For each assessor: documented training data, to exclude targets it may have seen.

Tools.

- Assessors: RNArank, lociPARSE, and the scoring functions and meta-scoring functions of RNAdvisor 2 (published November 2025; rsRNASP, DFIRE-RNA, RASP and others); ARES if it runs.
- Consensus: pairwise TM-score among models (US-align) and 2D agreement (RNApdbee or barnaba).
- Ground truth: RNAdvisor for lDDT, TM-score and INF against the reference.

Metrics.

- Per-target Spearman correlation between each signal and true lDDT or TM-score.
- Selection regret = quality of the best available model − quality of the chosen model (zero when the assessor picks the best, positive otherwise), per target and averaged, with target-level bootstrap intervals.
- AUC for separating good models (TM-score > 0.5) from the rest; results with and without potentially contaminated targets.
- Combination methods compared head to head: RNAdvisor 2’s published meta-scoring functions against a training-free rank average of the best single signals, and both against the best single signal.

Week-by-week plan (item number = week).

1. Roles; download CASP16 server models; install RNAdvisor; score ten models.
2. Collect RNA-Puzzles models; document each assessor’s training data from papers and repositories; refresh the closest prior work.
3. Freeze the protocol: pools, assessors, metrics (including the regret definition), contamination exclusions, tool versions.
4. Compute true quality of all public models (lDDT, TM-score, INF).
5. Run RNAdvisor 2 scoring functions and meta-scoring functions on all models.
6. Run RNArank and lociPARSE on all models.
7. Compute consensus scores; parse predictor confidence where available.
8. MVP (required): ranking statistics and regret on CASP16 pools; internal review.

9. Main extension: own multi-predictor pool on 30–50 targets (Boltz-2 and Chai-1 first); score everything.
10. Analysis: regret, per-target correlation, by target difficulty; cross-predictor pools.
11. Combination: RNAdvisor 2 meta-scoring functions versus rank average versus the best single signal.
12. Stretch: case studies of targets where confidence misleads.
13. Robustness: bootstrap intervals, contamination exclusion, length effects.
14. Freeze results; make figures.
15. Draft the report; clean the code.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
An assessor is unavailable or broken	Use the scoring functions bundled in RNAdvisor 2.
Contamination cannot be resolved	Report both with and without the doubtful targets.
CASP16 pools not available in bulk	RNA-Puzzles pools plus own multi-predictor pools.
No GPU for own pools	Boltz-2 only, fewer seeds; public pools carry the study.

Scope.

Class	Work
Required (MVP)	Public CASP16 and RNA-Puzzles pools: predictor confidence, assessors, consensus and RNAdvisor 2 meta-scores, with regret and contamination documented.
Main extension	Own multi-predictor pools on 30–50 targets.
Stretch	Combination tuning; case studies.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	RNArank (2026); lociPARSE; RNAdvisor 2 meta-scoring functions (November 2025); CASP16 model-quality analyses.
Specific new contribution	Assessors, meta-scores and predictor confidences compared on the practical task of choosing among models from several predictors, with regret and contamination made explicit.
Main endpoint	Mean selection regret (lDDT) per signal, with target-level intervals.
Evaluation risks	Assessor training overlap with CASP15/16 targets; pools unavailable; confidence scales not comparable across predictors.
Feasibility evidence	CASP16 server models and RNAdvisor running in week 1.

Brief 7 — Calibration of base-pair probabilities, and their relation to chemical probing

Difficulty 2 · Recommended team 2–3 · CPU · MVP (week 8): calibration of three predictors against base-pair annotations from a family-disjoint set of experimental structures, plus the reactivity association on 20,000 Ribonanza sequences, the two reported separately.

Question. Are the marginal base-pairing probabilities from RNA secondary-structure predictors calibrated against experimental base-pair annotations on a family-disjoint structural test set? Separately, how well do these probabilities relate to probe-specific chemical reactivity (DMS, 2A3), and does that relationship transfer across sequence sources? Chemical reactivity is treated throughout as an experimental observation that depends on the probe, nucleotide identity, local environment and dynamics, never as a direct label of paired versus unpaired.

Data.

- Calibration ground truth: base-pair annotations from experimental structures (PDB-derived, extracted with RNApdbee or barnaba), clustered by Rfam family so that test families are disjoint from any training family of the learned predictors; pseudoknot handling fixed in the protocol.
- Association data: Ribonanza chemical mapping (2A3 SHAPE and DMS) for about two million sequences; the high signal-to-noise subset, subsampled to 20,000–50,000 sequences, split by sequence cluster and labelled by source (designed versus natural); mutate-and-map subsets for the stretch analysis.
- Optional: an in-cell DMS-MaPseq dataset for an out-of-distribution check.

Tools.

- Predictors with pair probabilities: ViennaRNA (RNAfold -p), LinearPartition, CONTRAfold, EternaFold, RNAstructure, MXfold2; SPOT-RNA and UFold where probability outputs exist; RibonanzaNet as a reactivity-prediction reference.
- Analysis: Python with numpy and scikit-learn-style calibration code (reliability diagrams, isotonic regression, temperature scaling), cluster-level bootstrap.

Metrics.

- Calibration (structural ground truth): P(paired) per nucleotide = Σ pair probabilities; reliability diagrams, expected calibration error, Brier score and log loss per predictor; post-hoc temperature scaling fitted on held-out families.
- Association (reactivity): P(unpaired) = 1 − P(paired) against continuous reactivity, per probe (DMS on A/C only; 2A3 on all nucleotides): Spearman correlation and a regression fit, with any threshold-based classification metric reported only as secondary; transfer measured by fitting on one sequence source and testing on another.
- Stratification by sequence source and length; cluster-level bootstrap for all intervals.

Week-by-week plan (item number = week).

1. Roles; install ViennaRNA, LinearPartition, CONTRAfold, EternaFold; download the Ribonanza subset; smoke test on ten sequences.

2. Build the structural ground truth: PDB-derived pairs, family labels, family-disjoint split; refresh the closest prior work, including the 2026 generalizability study by Szikszai and colleagues.
3. Freeze the protocol: both ground truths, metrics, splits, predictors, tool versions.
4. Harness: run predictors → pair-probability matrices → P(paired); sparse storage.
5. Run ViennaRNA and LinearPartition on both datasets (CPU batch).
6. Run CONTRAfold, EternaFold, MXfold2; add SPOT-RNA or UFold if probabilities are available.
7. Calibration analysis on the structural set; association analysis on Ribonanza.
8. MVP (required): both analyses for three or more predictors, reported separately; internal review.
9. Main extension: post-hoc calibration (temperature scaling, isotonic) fitted on held-out families; transfer of the reactivity relation across sequence sources.
10. Analysis by structural context: helix ends, loops, pseudoknots.
11. Stretch: in-cell DMS-MaPseq check.
12. Stretch: mutate-and-map: do probability changes under mutation track reactivity changes?
13. Robustness: signal-to-noise filter, cluster-level resampling, canonical-only versus all annotated pairs.
14. Freeze results; make figures.
15. Draft the report; clean the code.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
Reactivity read as pairing status	By design it never is: the two analyses stay separate and the report states what each one measures.
Structural set too small for some families	Merge rare families into one held-out group; report per-family counts.
Data volume strains laptops	Subsample further; run on a shared CPU node.
A predictor gives no probabilities	Sample structures (RNAsubopt -p) to estimate them, or exclude.

Scope.

Class	Work
Required (MVP)	Calibration on the family-disjoint structural set and the reactivity association on Ribonanza, three or more predictors, reported separately.
Main extension	Post-hoc calibration on held-out families; transfer of the reactivity relation across sequence sources.
Stretch	In-cell probing data; mutate-and-map.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	Szikszai et al., 2022 (family generalization) and 2026 (generalizability benchmark, prediction of probing data); Ribonanza and RibonanzaNet; EternaFold (trained on probing data); CHANRG (2026).
Specific new contribution	Calibration, not accuracy, of pair probabilities on a family-disjoint

Field	Starting point
	structural set, plus a separate, probe-specific account of how probabilities relate to reactivity.
Main endpoint	Expected calibration error per predictor on the family-disjoint structural set.
Evaluation risks	Pseudoknot conventions, training overlap between learned predictors and the structural set, probe-specific biases.
Feasibility evidence	Structural set built in week 2; all predictors run on CPU.

Brief 9 — MSA ablation for RNA co-folding

Difficulty 3 · Recommended team 3–4 · CPU heavy, then GPU · MVP (week 8): one predictor that documents consuming an RNA alignment (RhoFold+ or trRosettaRNA; Protenix only if its alignment route passes the week-1 test), run on 40 Rfam-covered targets with no alignment versus Rfam alignments subsampled to four depths, with Boltz-2 and Chai-1 as single-sequence comparators.

Question. How much of a 3D predictor's RNA accuracy comes from the multiple sequence alignment, and how do alignment source and depth change the result? Secondary question: does the model's confidence rise with alignment depth even when accuracy does not? Only predictors that document consuming an RNA alignment enter the depth series; the ability to build an alignment is not evidence that a tool uses it.

Data.

- Targets: 40–60 RNA monomers (40–200 nt) with an Rfam family of at least 100 sequences and an experimental structure, post-cutoff where possible, otherwise drawn from RNA3DB-style test clusters.
- Alignments: (a) none; (b) the Rfam full alignment subsampled to depths of 1, 10, 100 and 1,000 with effective depth (Neff) recorded; (c) rMSA; (d) RNAcmap3 or an elDORS-augmented search if the database is accessible; (e) synthetic homologs if the tool is released.
- Alignment route verified in week 1 for every predictor in the series: its documented input (A3M or its own format) and a test showing that a change in alignment depth changes the output.

Tools.

- Predictors with a documented RNA alignment input: RhoFold+ and trRosettaRNA (RNA-specific, consume an alignment directly), RNAPro (alignment and templates as inputs), AlphaFold 3 if weights are available. Protenix (v1.0.0 and later) documents an RNA alignment search (`--use_rna_msa`) but not a custom alignment input, so it joins the depth series only if the week-1 test shows that a substituted alignment is consumed; otherwise it contributes an alignment on/off comparison only.
- Single-sequence comparators: Boltz-2 (its custom alignment field is protein-only) and Chai-1, run without alignments, to show where the single-sequence frontier sits.
- Alignment building: Infernal (cmalign against the Rfam covariance model), rMSA, MMseqs2; a subsampling script (random and diversity-maximizing) and a Neff calculator; conversion to each tool's format.
- Evaluation: US-align, RNAdvisor.

Metrics.

- TM-score, lDDT and INF per condition; per-target gain (alignment minus single sequence); slope of accuracy against log Neff, per predictor and never pooled across predictors with different alignment pipelines.
- Comparison of sources at matched depth; confidence (pLDDT) against depth and against actual gain.
- Disk and CPU cost per alignment source; share of targets with any gain.

Week-by-week plan (item number = week).

1. Roles; install RhoFold+ (or trRosettaRNA) and Infernal; smoke test on two targets with and without an alignment, confirming that a changed alignment changes the output; runtime recorded.
2. Build the target set; download Rfam full alignments and covariance models; test Protenix's alignment route; refresh the closest prior work.
3. Freeze the protocol: predictors, conditions, depths, metrics, hypotheses, tool versions.
4. Alignment pipeline: cmalign, subsampling, Neff, format conversion; validate on five targets.
5. Run the primary predictor single-sequence and over the Rfam depth series (3 seeds).
6. Evaluation harness; first accuracy-against-depth curves.
7. Run Boltz-2 and Chai-1 single-sequence on the same targets; start rMSA runs on a cluster (CPU heavy; runs through week 9).
8. MVP (required): depth series for the primary predictor on 40 targets, with the single-sequence comparators; internal review.
9. Main extension: a second alignment-consuming predictor (trRosettaRNA, RNAPro or Protenix) under the same conditions.
10. Main extension continued: rMSA alignments at matched depth.
11. Analysis: which targets gain, and how that relates to family size, length and pseudoknots.
12. Stretch: elDORS-augmented or synthetic-homolog alignments; confidence analysis against depth.
13. Robustness: seeds; random versus diversity-maximizing subsampling.
14. Freeze results; make figures.
15. Draft the report; clean the code.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
No predictor's alignment route passes the week-1 test	Decided by week 2: switch to an alignment on/off comparison with Protenix's documented search, or to a single-sequence-versus-template question; the depth series is dropped, not faked.
rMSA databases too large for local disk	Infernal searches against RNAcentral only, or the Rfam series alone.
Families too small for the depth series	Choose families with at least 100 sequences; drop the 1,000 level if needed.
GPU time short	One seed per condition; fewer targets.

Scope.

Class	Work
Required (MVP)	Depth series on one alignment-consuming predictor, 40 targets, plus single-sequence comparators.
Main extension	A second alignment-consuming predictor; rMSA alignments at matched depth.
Stretch	elDORS or synthetic homologs; confidence-versus-depth analysis.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	rMSA and RNAcmap3 (alignment effects on 2D and contact prediction); elDORS (2026, alignment depth and 2D accuracy); the synthetic-homolog preprint (2026, AlphaFold 3 alignment dependence); custom alignments used by CASP16 teams.
Specific new contribution	A controlled depth series measured on 3D accuracy, with the single-sequence frontier of 2026 co-folding models as the reference.
Main endpoint	Slope of TM-score against log Neff for the primary predictor, with intervals.
Evaluation risks	Tools that silently ignore alignments, Rfam family overlap with training data, depth confounded with family size.
Feasibility evidence	Week-1 proof that the predictor consumes the alignment.

Brief 10 — Leakage audit of RNA language-model benchmarks

Difficulty 2 · Recommended team 3–4 · CPU (GPU optional) · MVP (week 8): an overlap audit, in four categories, of BEACON test sets against two models' documented proxy pretraining corpora and the benchmark's own train splits; the audit is a complete deliverable on its own.

Question. How much of the test data in widely used RNA language-model benchmarks is near-identical, by sequence or by Rfam family, to the models' pretraining corpora or to the benchmarks' own training splits, and does measured performance drop on the non-overlapping part?

Data.

- Benchmarks: BEACON (13 tasks), RNA-Scope (15 core subtasks, with its own generalization analyses across families and contexts), and common secondary-structure sets (bpRNA, ArchiveII, bpRNA-new).
- Pretraining corpora, reconstructed per model from papers and repositories; they differ (RNA-FM reports about 23.7 million RNAcentral sequences; RiNALMo about 36 million from RNAcentral, nt, Rfam and Ensembl), so each model gets its own documented proxy corpus with release version, and no model is assumed to share another's corpus.
- Rfam for family assignment; published fine-tuned checkpoints for the re-evaluation step.

Tools.

- MMseqs2 (easy-search and clustering at 80, 90 and 95% identity with coverage thresholds), CD-HIT-est or BLASTN for small sets, Infernal cmscan for families.
- Model inference from published checkpoints and the benchmarks' own evaluation scripts; Python for overlap statistics.

Metrics.

- Four overlap categories, each reported separately: exact sequence overlap; near-identical overlap at pre-registered identity and coverage thresholds (90% and 80% identity); Rfam-family overlap; unknown or incompletely documented training exposure.
- Share of test sequences in each category, per task and per model, against the proxy corpus and against the benchmark's own train split; labelled as measured overlap with a documented proxy corpus whenever the exact training set cannot be reconstructed, never as proof of leakage.
- Performance on clean versus overlapping subsets for re-evaluated models, with effect sizes; a per-task leakage map; threshold sensitivity.

Week-by-week plan (item number = week).

1. Roles; download BEACON datasets; install MMseqs2 and Infernal; collect model and corpus documentation.
2. Corpus inventory: which release and which sources per model; download the releases; refresh the closest prior work (BEACON, RNA-Scope, the 2026 zero-shot benchmark).

3. Freeze the protocol before any search: identity and coverage thresholds, the family criterion, the four categories, tasks and models to re-evaluate.
4. Test-versus-train overlap within each benchmark task.
5. Test-versus-proxy-corpus overlap for each model (cluster first, then search).
6. Family assignment for all test and train sequences; family overlap statistics.
7. Build clean subsets per task; data tables by category.
8. MVP (required): the four-category overlap map for BEACON tasks against two proxy corpora; internal review.
9. Main extension: re-evaluate one model's checkpoints on clean versus overlapping subsets for three tasks.
10. Main extension continued: a second model, or RNA-Scope tasks.
11. Analysis: tasks where overlap matters most; relation to reported gains over baselines.
12. Stretch: package the cleaned split files as a reusable resource.
13. Robustness: threshold and coverage sensitivity.
14. Freeze results; make figures.
15. Draft the report; clean the code.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
Corpus version undocumented	Use the latest release before the paper's submission date; state the assumption.
RNAcentral too large for local search	Cluster at 90% first on a shared node, or search test sets against the corpus rather than the reverse.
Checkpoints unavailable	Report the overlap statistics alone; they stand on their own.
Benchmark scripts broken	Re-implement the published metric and validate on a released number.

Scope.

Class	Work
Required (MVP)	Four-category overlap audit of BEACON test sets against two documented proxy corpora and the train splits.
Main extension	Re-evaluation of published checkpoints on clean versus overlapping subsets.
Stretch	RNA-Scope tasks; release of cleaned split files.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	BEACON (2024); RNA-Scope (2025); the 2026 zero-shot evaluation of 21 RNA language models; CHANRG (2026); Szikszai et al., 2022.
Specific new contribution	A reproducible, category-resolved overlap audit between evaluation sets and documented proxy pretraining corpora, which none of the benchmarks report.
Main endpoint	Share of test sequences per overlap category per task, and the clean-versus-overlap performance gap where checkpoints run.
Evaluation risks	Corpus reconstruction, threshold choice, broken benchmark scripts.
Feasibility evidence	An RNAcentral release downloaded and clustered on a shared node by week 2.

Brief 11 — Replication study of a recent RNA machine-learning paper

Difficulty 3 (2–4 depending on the paper) · Recommended team 2–4 · Compute depends on the paper · MVP (week 8): a claimed-versus-obtained table for one headline claim plus an audit of the paper's train/test splits; one faithfully reproduced claim with an independent test set beats several incomplete ones.

Question. Do the headline results of a recent RNA machine-learning paper reproduce with its released code and weights, and do they hold on a test set that is newer (temporal) or family-disjoint from its training data?

Data.

- Candidate papers (chosen in week 2 from a shortlist of five): recent models with released weights and data in secondary-structure prediction, RNA language models, RNA 3D prediction or RNA design.
- The paper's own test sets and reported numbers (tables and figures to reproduce).
- A new test set built by the team: PDB entries released after the paper's data cutoff, clustered RNA3DB-style, or Rfam families absent from the training data.

Tools.

- The paper's repository in a pinned environment (conda or Docker); its evaluation scripts.
- Independent re-implementation of the metrics (base-pair F1 with and without one-nucleotide shifts, TM-score, lDDT, task-specific scores); MMseqs2 and Infernal for split audits; RNAdvisor for 3D tasks.

Metrics.

- Reproduction: absolute and relative difference between obtained and reported numbers, against the paper's own variance where reported.
- Split audit: sequence identity and family overlap between the paper's train and test sets.
- Generalization: performance on the new test set against the baselines the paper itself names (for example ViennaRNA for 2D); a numbered list of discrepancies between code and paper; resource and runtime reproducibility.

Week-by-week plan (item number = week).

1. Roles; shortlist five papers; check code, weights and data availability for each.
2. Choose the paper; build the environment; run the smoke test on the released weights and data.
3. Go/no-go: environment runs, weights load, data accessible; otherwise switch to the second candidate now. Freeze the protocol: the one headline claim to reproduce, tolerance, new test-set rules, tool versions.
4. Reproduce inference for the headline claim on the paper's test set; compare numbers.
5. Re-implement the metric independently; cross-check against the paper's scripts.
6. Audit the paper's splits (identity, family overlap).
7. Build the new temporal or family-disjoint test set.
8. MVP (required): claimed-versus-obtained for the headline claim plus the split audit; internal review.

9. Main extension: the new test set, with the paper's named baselines.
10. Stretch: a second claim, or retraining if cheap; otherwise the ablations the code supports.
11. Write to the authors with precise, courteous questions on discrepancies; fold in replies.
12. Analysis: which claims hold, which do not, and the failure modes.
13. Robustness: seeds, library versions, hardware differences.
14. Freeze results; make figures.
15. Draft the report as a research article, with results organized as claim → outcome → deviation.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
The code will not run	Go/no-go in week 3: document the failure precisely (that is a result) and switch to the second candidate.
Retraining is too costly	Inference-only replication plus the new test set.
Authors do not answer	Report the discrepancies as open questions.
Everything reproduces exactly	The new test set supplies the novel part of the study.

Scope.

Class	Work
Required (MVP)	One headline claim reproduced from released code and weights, plus the split audit.
Main extension	The new temporal or family-disjoint test set with the paper's named baselines.
Stretch	A second claim, or retraining where it is cheap.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	The chosen paper and the baselines it names; any existing reproduction or benchmark that includes it.
Specific new contribution	An independent reproduction plus a test the paper did not run.
Main endpoint	Obtained-minus-claimed difference for the headline metric, and the same metric on the new test set.
Evaluation risks	Environment drift, undocumented preprocessing, author data unavailable, test-set overlap with the paper's training data.
Feasibility evidence	Go/no-go passed in week 3.

Brief 12 — Systematic review of evaluation practice in RNA machine learning

Difficulty 2 (high labor, low technical risk) · Team 4–6 · No compute · MVP (week 8): prevalence table and annotator agreement on the first 100 fully annotated papers.

Question. What share of RNA machine-learning papers from 2022 to a fixed search cutoff date evaluate with family-level or temporal splits, check train/test leakage, report confidence intervals and release code and data, and has practice changed since the 2022 warning that deep-learning secondary-structure predictors may not generalize across families (Szikszai et al., 2022, revisited in their 2026 generalizability study)?

Data.

- Records retrieved through the OpenAlex and Europe PMC APIs plus preprint servers, with pre-registered search strings and a fixed search end date set in week 3; 2026 is reported as a partial year and never compared with full years without that label.
- Scope: RNA secondary and 3D structure prediction, RNA language models, RNA–protein and RNA–ligand interaction, modification-site prediction, RNA design; the search protocol explicitly includes the 2025–2026 generalization literature (RNA-Scope, CHANRG, the 2026 generalizability study).
- Sample capped for feasibility: at most 300 papers after screening (a stratified random sample if more), so that full-text double annotation and adjudication fit the semester.
- A codebook with fields such as split type, leakage check, baselines, confidence intervals, dataset names, code and data availability, reported gain over the best baseline.

Tools.

- Retrieval and deduplication scripts (Python); a shared screening sheet or a reference manager; a PRISMA-style flow diagram.
- Statistics in Python or R: prevalence with 95% intervals, logistic trend by year, Cohen's kappa for annotator agreement.

Metrics.

- Prevalence of each practice with intervals; trend by year; differences by subfield.
- Annotation reliability (kappa) and adjudication rate.
- Association between leakage control and the size of the reported improvement over baselines.

Week-by-week plan (item number = week).

1. Roles; define scope and research questions; draft search strings.
2. Pilot search; estimate volume; refine strings and inclusion criteria; refresh the closest prior work.
3. Freeze the protocol: inclusion and exclusion criteria, codebook, search end date, sample cap, double-annotation rule.
4. Retrieve records; deduplicate; start title and abstract screening (two annotators per record).
5. Finish screening; draw the capped sample; resolve disagreements; compute kappa.
6. Codebook calibration: all annotators code the same 20 papers, discuss, refine.

7. Full-text annotation, batch 1 (each paper by two annotators).
8. MVP (required): prevalence table and kappa on 100 papers; internal review.
9. Main extension: annotation batch 2.
10. Main extension continued: batch 3; adjudicate disagreements.
11. Analysis: prevalence, trends with 2026 marked partial, subfield differences.
12. Stretch: association analysis, leakage control versus claimed gains.
13. Robustness: sensitivity to inclusion criteria; annotator effects.
14. Freeze results; make the flow diagram and trend figures.
15. Draft the report; release the annotated dataset with the codebook.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
Volume too large for the team	The 300-paper cap with a documented stratified random sample.
Codebook ambiguities	Pilot on 20 papers in week 6 and revise before batch 1 finishes.
Low annotator agreement	Written adjudication rules; a third annotator breaks ties.
Annotator drift over weeks	Re-calibration on five shared papers at the start of each batch.
The current year is incomplete	2026 labelled partial in every figure; trend tests exclude it or model it as such.

Scope.

Class	Work
Required (MVP)	100 papers double-annotated, prevalence table and agreement statistics.
Main extension	The full capped sample with trends by year and subfield.
Stretch	Association analyses and sensitivity to criteria.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	Szikszai et al., 2022 and 2026; CHANRG (2026); RNA-Scope (2025); evaluation audits in other areas of machine learning for biology.
Specific new contribution	A quantified, annotated record of evaluation practice across RNA machine-learning subfields, with a reusable codebook and dataset.
Main endpoint	Prevalence, with intervals, of family-level or temporal splits per year.
Evaluation risks	Annotation reliability, inclusion criteria, the incomplete current year.
Feasibility evidence	Pilot search volume measured in week 2.

Brief 14 — A harmonized, confidence-tiered evidence atlas of RNA modification sites

Difficulty 3 · Recommended team 3–4 · No compute · MVP (week 8): an evidence atlas for m6A and pseudouridine in HEK293T, with per-site assay provenance, evidence tiers and method-agreement statistics computed after coverage matching.

Question. Can published site-level calls from orthogonal RNA-modification assays be harmonized into a provenance-rich, assay-aware evidence atlas for a defined cellular context, and how does agreement between methods change after matching coverage and other quality criteria? Support by independent methods is treated as evidence strength, never as proof that a site is real: the atlas is a record of evidence, not a ground truth.

Data.

- Cell lines with several orthogonal datasets (HEK293T first, HeLa second): m6A (GLORI, eTAM-seq, m6A-SAC-seq, miCLIP), pseudouridine (BID-seq, PRAISE, RBS-seq), m5C (bisulfite-based methods), inosine (editing databases such as REDIportal), 2′-O-methylation (Nm-seq, RibOxi-seq).
- Context rules fixed in week 3: which datasets count as the same cell line, growth condition and RNA fraction, and which quality measures (coverage, replicate support, stoichiometry) must be matched before agreement is computed. The National Academies review of RNA sequencing and modifications documents how much protocols, sample preparation and analysis vary between studies, and calls for orthogonal validation and standardization.
- Supplementary tables from the source papers; GENCODE annotation; genome builds hg19 and hg38.
- Existing databases as comparators, not as sources: RMBase v3.0, m6A-Atlas, REDIportal. The atlas must state what it adds: per-site assay and study provenance, context matching, coverage-matched agreement and explicit evidence tiers.

Tools.

- Python with pandas; liftOver or CrossMap for coordinate conversion; bedtools; a validation layer (pandera or similar) with unit tests.
- A fixed schema per site: coordinate (hg38), strand, modification, assay, study and DOI, replicate, cell line and condition, stoichiometry, coverage, original coordinate system, transcript provenance.
- Export as BED and TSV with a data dictionary; versioned releases traceable to the original studies.

Metrics.

- Pairwise site-level agreement between methods (Jaccard) computed only after coverage and quality matching within the same context, per modification; stoichiometry correlation where both methods report it.
- Motif consistency (for example DRACH for m6A); coordinate-conversion loss rates.
- Evidence tiers: T1 = at least two orthogonal methods in a matched context, T2 = one high-confidence method, T3 = a single lower-confidence method; counts per tier and discrepancies with existing databases.

Week-by-week plan (item number = week).

1. Roles; inventory datasets per modification and cell line, with context metadata; collect supplementary tables.
2. Define the schema and the tiers; draft merging and context-matching rules; refresh the closest prior work (RMBase v3.0, m6A-Atlas, the National Academies review).
3. Freeze the protocol: merging rules, context rules, coverage thresholds, agreement metrics, exclusions.
4. Parse m6A datasets into the schema; liftOver to hg38; validation tests.
5. Parse pseudouridine datasets into the schema.
6. Agreement analysis for m6A and pseudouridine after coverage matching; motif checks.
7. Build the merged, tiered atlas for HEK293T; run validation.
8. MVP (required): m6A and pseudouridine atlas for HEK293T with agreement statistics; internal review.
9. Main extension: m5C and inosine; editing-database conventions.
10. Main extension continued: HeLa as a second context.
11. Cross-check against existing databases; document discrepancies and what the atlas adds.
12. Stretch: 2′-O-methylation; stoichiometry harmonization where methods allow, otherwise flagged as not comparable.
13. Robustness: coverage thresholds, replicate requirements, context-rule sensitivity.
14. Freeze the release; make agreement heatmaps and tier-count figures.
15. Draft the report; write the data dictionary and provenance notes.
16. Final report, repository and versioned data release, presentation.

Risks.

Risk	Fallback
Supplementary tables incomplete or inconsistent	Write to the authors; log exclusions per dataset.
Transcript versus genome coordinates	Map everything to hg38 and keep the original coordinate system and transcript provenance in columns.
Datasets from different contexts merged as if comparable	The context rules exclude them from agreement statistics; they can still be listed with their provenance.
Methods disagree strongly	That is a finding; the tiers make the disagreement visible.
The atlas is read as ground truth	The title, the tier definitions and the documentation say evidence; every site keeps its assay provenance.
Scope creep	Two modifications and one cell line until week 8.

Scope.

Class	Work
Required (MVP)	m6A and pseudouridine in HEK293T, with provenance, context matching, coverage-matched agreement and tiers.
Main extension	m5C and inosine; HeLa as a second context.
Stretch	2′-O-methylation; stoichiometry harmonization.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	RMBase v3.0; m6A-Atlas; REDIportal; the 2025 RNA004 modification-calling benchmark; the National Academies review of RNA-modification standards.
Specific new contribution	Coverage-matched, assay-aware agreement statistics and explicit evidence tiers with full provenance for one cellular context, which existing databases do not provide.
Main endpoint	Jaccard agreement between orthogonal methods per modification after coverage matching, and the tier counts.
Evaluation risks	Context mismatch, coordinate conversion, incomplete supplementary data, tiers read as truth.
Feasibility evidence	Dataset inventory with context metadata completed by week 2.

Brief 15 — Rfam's structural dark matter

Difficulty 3 · Recommended team 3–4 · GPU moderate · MVP (week 8): a versioned catalogue, tied to Rfam 15.1, of families under 150 nt with no known experimental structure under a documented search procedure, each classified by prediction confidence and consistency, calibrated on families with known structures.

Question. For Rfam families with no experimental 3D structure known under a documented search procedure, which can be modeled confidently and consistently by current predictors, which cannot, and where do predicted 3D structures contradict the covariance-supported consensus secondary structure? The deliverable is a versioned catalogue of predicted confidence and consistency, not a claim that any family's fold is solved.

Data.

- Rfam 15.1 (January 2026; 4,227 families), pinned with its release date; the candidate list of families without an experimental structure is recomputed at project start, never copied from an earlier release.
- A documented structure search per candidate family at family and domain level: the Rfam PDB mapping plus an Infernal search of the family model against PDB RNA sequences. A family is labelled as having no known experimental structure under this procedure, never as having none at all.
- One or two representative sequences per family (seed sequence closest to the consensus), under 150 nt for the MVP and under 300 nt afterwards.
- Positive controls: families with experimental structures, excluded from the target set, used to calibrate confidence and consistency thresholds; the limits of that calibration (control families are better studied and often shorter) are reported.

Tools.

- Predictors: Boltz-2 (primary); Chai-1 or Protenix on a subset for cross-tool consistency.
- Infernal for representative selection and alignment-column mapping; RNApdbee or barnaba for secondary structure from 3D models; ViennaRNA for a sequence-only baseline; US-align for pairwise model comparison; RNAdvisor for controls.

Metrics.

- Confidence: pLDDT distribution per family; consistency: pairwise TM-score across seeds and across tools.
- Agreement with Rfam: base-pair F1 between the consensus structure (mapped to the representative) and the pairs extracted from the model.
- Family categories: confident and consistent; confident but contradicting Rfam; low confidence. Labels describe predicted confidence and consistency, never demonstrated structural correctness. Recovery on controls (TM-score to the PDB structure); category frequencies by family type (riboswitch, sRNA, snoRNA, cis-regulatory element, lncRNA domain).

Week-by-week plan (item number = week).

1. Roles; download Rfam 15.1; recompute the list of families without PDB mapping; smoke test with Boltz-2.

2. Run the domain-level structure search for candidate families; representative-selection pipeline; map consensus structures onto representatives; refresh the closest prior work.
3. Freeze the protocol: search procedure, metrics, categories, thresholds, control set, tool versions, the pinned release.
4. Harness: predict → extract 2D → compare with consensus; consistency across seeds.
5. Run Boltz-2 on the positive controls; set thresholds from their distributions; record the calibration limits.
6. Run Boltz-2 on all target families under 150 nt (3 seeds).
7. Compute metrics; first categorization.
8. MVP (required): catalogue for families under 150 nt with calibrated categories; internal review.
9. Main extension: second predictor on a stratified subset; cross-tool consistency.
10. Main extension continued: families under 300 nt as compute allows.
11. Stretch: manual inspection of 20 candidate novel folds; which family types are modeled confidently.
12. Stretch: contradictions with Rfam, such as pseudoknots, alternative structures, alignment artefacts.
13. Robustness: second representative per family; seeds.
14. Freeze results; make figures and a sortable catalogue table.
15. Draft the report; package the versioned catalogue.
16. Final report, repository and catalogue release, presentation.

Risks.

Risk	Fallback
Thousands of families exceed the GPU budget	Prioritize by length and family size; one seed beyond the MVP set.
Consensus mapping errors	Validate the mapping on the control families before the batch.
A family labelled structure-free turns out to have a structure	The label is procedural; the search procedure is published with the catalogue and later finds are noted, not hidden.
Over-claiming novel folds	Report confidence categories only; no functional claims without experimental support.
Rfam release changes mid-project	Pin 15.1 in week 1; state the release in every table; mention newer releases at the end.

Scope.

Class	Work
Required (MVP)	Families under 150 nt, Boltz-2, calibrated categories, documented search procedure, Rfam 15.1 pinned.
Main extension	A second predictor; families under 300 nt.
Stretch	Novel-fold inspection; contradictions with Rfam; catalogue release.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	Rfam 15 and 15.1 (2025–2026); RNA3DB; the co-folding model papers; CHANRG (2026).
Specific new contribution	A versioned, calibrated catalogue of prediction confidence and

Field	Starting point
	consistency over structure-free Rfam families, with an explicit structure-search procedure.
Main endpoint	Share of candidate families in each category, with the calibration on controls.
Evaluation risks	Mapping gaps, representative choice, over-claiming, calibration limits of control families.
Feasibility evidence	Candidate list and control set computed by week 2.

Brief 16 — Independent evaluation of 3D RNA inverse-folding designs

Difficulty 4 · Recommended team 4–5 · GPU heavy · MVP (week 8): gRNAde designs on at least 40 non-redundant target backbones, refolded with one independent predictor (Boltz-2), native sequences as reference controls, success rates reported as curves over thresholds.

Question. How do independently trained, contemporary RNA structure predictors change the estimated designability of sequences generated by an RNA inverse-folding model (gRNAde), and how consistent are structural success assessments across predictors on a structurally non-redundant test set? The gRNAde paper already reports native-sequence recovery and self-consistency with RhoFold, so repeating that loop is not the contribution; the new question is what an independent judge, and the agreement between judges, do to those conclusions.

Data.

- Target backbones: experimental RNA structures of 40–100 nt, non-redundant and held out from gRNAde's training data by documented sequence and structural exclusion rules (RNA3DB-style clustering), post-cutoff for the refolding predictors where possible; a few de novo scaffolds as a separate set.
- Native sequences as reference controls, not as an assumed ceiling: the native sequence is not guaranteed to be the sequence a predictor refolds best.
- Target secondary structures extracted from the backbones for the 2D check.

Tools.

- Design: gRNAde (16 sequences per target at low sampling temperature, matching its paper's protocol so results are comparable); RiboDiffusion only as stretch.
- Refolding: Boltz-2 as the primary independent predictor; Chai-1 or Protenix as the main extension; RhoFold, the paper's own judge, as a reference point.
- Folding engines for the 2D check: ViennaRNA, EternaFold, LinearFold.
- Evaluation: US-align (self-consistency TM-score and RMSD), RNAdvisor; sequence metrics (recovery, GC content, diversity).

Metrics.

- Per target: self-consistency TM-score and RMSD of each design, with the native-sequence refold reported beside them; target-level success rates with bootstrap intervals.
- Threshold sensitivity curves: success rate as a continuous function of the TM-score and RMSD cut-offs, with the gRNAde paper's choices (TM-score ≥ 0.45, RMSD ≤ 2 Å, GDT ≥ 0.5) marked as one point among many, never as the correct threshold.
- Agreement between predictors: Cohen's kappa on success calls and correlation of self-consistency scores; the share of targets where the verdict flips between judges.
- Sequence recovery against the native sequence; 2D agreement (base-pair F1) per folding engine; success by target feature (length, pseudoknots, non-canonical pair content).

Week-by-week plan (item number = week).

1. Roles; install gRNAde and Boltz-2; smoke test on one target, runtime and memory recorded.

2. Build the target set with documented exclusion rules; extract target secondary structures; refresh the closest prior work, starting from gRNAde's own evaluation.
3. Freeze the protocol: metrics, threshold ranges, designs per target, exclusion rules, tool versions.
4. Harness: design → refold → compare; test on five targets.
5. Pilot: eight designs per target on 20 targets, refolded with Boltz-2; native controls; set the compute budget from the measured runtimes.
6. Scale to 16 designs per target on 40 or more targets; refold with Boltz-2 (one seed).
7. Self-consistency metrics; threshold sensitivity curves; native controls.
8. MVP (required): gRNAde × Boltz-2 on 40 or more targets with controls and sensitivity curves; internal review.
9. Main extension: Chai-1 or Protenix refolding; agreement analysis; RhoFold as the reference judge.
10. Main extension continued: targets where the verdict flips; relation to target features.
11. Stretch: 2D folding-engine check; RiboDiffusion designs on the same targets.
12. Analysis: designability by target feature; where judges disagree.
13. Robustness: seeds, sampling temperature, number of designs per target, sensitivity to the exclusion rules.
14. Freeze results; make figures.
15. Draft the report; clean the code.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
GPU cost (designs × predictors)	Eight designs per target, one seed, fewer targets; the pilot sets the budget.
Targets overlap predictor training (refolding inflated)	Documented exclusion rules; post-cutoff targets preferred; overlap reported and stratified.
Second predictor unavailable	Single-judge study: threshold curves, native controls and RhoFold as reference still stand.
Success thresholds are arbitrary	Full sensitivity curves rather than one cut-off.

Scope.

Class	Work
Required (MVP)	gRNAde × Boltz-2 on 40 or more non-redundant targets, native controls, threshold sensitivity curves.
Main extension	A second independent judge and the agreement analysis.
Stretch	2D folding-engine check; RiboDiffusion designs.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	gRNAde (16 designs per target, recovery, self-consistency with RhoFold, thresholds TM-score ≥ 0.45, RMSD ≤ 2 Å, GDT ≥ 0.5); RiboDiffusion; RNA-FrameFlow; Eterna100-V2 (folding-engine dependence in 2D design).
Specific new contribution	Independent, multi-judge designability estimates with threshold sensitivity on a non-redundant, training-excluded target set.
Main endpoint	Success rate under Boltz-2 at the paper's thresholds, and Cohen's

Field	Starting point
	kappa between judges.
Evaluation risks	Training overlap of judges with targets, judge bias, compute.
Feasibility evidence	Pilot on 20 targets completed by week 5 with measured runtimes.

Brief 17 — Probing RNA language-model embeddings for structural annotations

Difficulty 3 · Team 3 · GPU light (inference only) · MVP (week 8): linear probes for Leontis–Westhof pair classes on one language model, with family-level splits, baselines and control tasks.

Question. Do RNA language-model embeddings linearly encode structural information beyond canonical pairing, namely non-canonical base-pair classes, tertiary contacts and modification sites, and does that information survive for families the model never saw?

Data.

- Annotations from PDB structures: base pairs with Leontis–Westhof classes and stacking from barnaba or FR3D, A-minor and other tertiary motifs from the RNA 3D Motif Atlas, modified residues from the structures and MODOMICS for tRNAs.
- Label semantics fixed in week 3: a positive is an annotated interaction in a resolved structure; a negative is a nucleotide or pair in a resolved region where the annotation tool reports no interaction (an observed negative); unresolved or unannotated regions are excluded, never counted as negatives.
- Sequences grouped by Rfam family for family-held-out splits; shuffled sequences as negative controls; a per-nucleotide table and a per-pair table (positives plus observed negatives sampled at matched sequence separation).
- Overlap between probe sequences and each language model's documented pretraining corpus, measured as in brief 10 and reported beside the results.

Tools.

- Language models, inference only: RNA-FM and RiNALMo first; ERNIE-RNA, UTR-LM or SpliceBERT where relevant; all layers extracted.
- Probes: logistic regression and a small MLP (scikit-learn); control tasks with random label assignments; baselines: one-hot plus position, ViennaRNA pair probabilities, and optionally covariance from an alignment.

Metrics.

- AUPRC as the principal metric for the imbalanced classes, AUROC secondary; selectivity (probe score minus control-task score), all on family-held-out test sets.
- Baselines fixed in the protocol: one-hot plus position, ViennaRNA pair probabilities, and a family-aware nearest-neighbour baseline; layer-wise curves; linear versus MLP probe gap.
- Per-pair probes on concatenated or element-wise product embeddings; calibration of probe outputs.

Week-by-week plan (item number = week).

1. Roles; install RNA-FM and RiNALMo; extract embeddings for ten sequences.
2. Build the annotation dataset from the PDB (barnaba): pair classes, stacking, A-minor; map to sequences and Rfam families; refresh the closest prior work.
3. Freeze the protocol: classes, label semantics (observed negatives), splits, probes, baselines, control tasks, tool versions.

4. Family-level splits; class-balance analysis; observed-negative sampling for pair probes; corpus-overlap check.
5. Extract embeddings (all layers) for the whole dataset; storage plan.
6. Nucleotide-level probes (modification sites, participation in non-canonical pairs) with baselines.
7. Pair-level probes for the main Leontis–Westhof classes.
8. MVP (required): probe results for one model and the pair classes, with baselines and controls; internal review.
9. Main extension: second language model; layer-wise analysis.
10. Main extension continued: analysis of what is encoded, in which layers, and whether it transfers across families.
11. Stretch: tertiary contacts (A-minor, long-range stacking) and modification sites.
12. Stretch: case study on a ribozyme with well-characterized tertiary contacts.
13. Robustness: seeds, probe capacity, class rebalancing, exclusion of sequences overlapping the pretraining corpus.
14. Freeze results; make figures.
15. Draft the report; clean the code.
16. Final report, repository, presentation.

Risks.

Risk	Fallback
Annotation-tool licensing	barnaba and FR3D-style open annotations only.
Unannotated treated as negative	The observed-negative rule; unresolved or unannotated regions are excluded from both classes.
Severe class imbalance	AUPRC as the headline metric; stratified splits; matched negatives.
Murky results	Control tasks and baselines make a null result interpretable; report it.
Large models exceed GPU memory	Batch on a shared node or restrict to smaller models.

Scope.

Class	Work
Required (MVP)	One language model, the main Leontis–Westhof pair classes, observed-negative labels, family-held-out splits, baselines and control tasks.
Main extension	A second language model; layer-wise analysis; transfer across families.
Stretch	Tertiary contacts and modification sites; a ribozyme case study.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	RNA-FM and RiNALMo papers (attention and canonical pairing); probing studies on protein language models; the structural tasks of BEACON and RNA-Scope; the 2026 zero-shot language-model benchmark.
Specific new contribution	Probing for non-canonical pair classes and tertiary contacts with family-held-out splits, observed-negative labels and corpus-overlap reporting.
Main endpoint	AUPRC per Leontis–Westhof class for the best layer, against the best baseline.

Field	Starting point
Evaluation risks	Label noise, imbalance, pretraining-corpus overlap, probe capacity.
Feasibility evidence	Annotation pipeline producing at least 5,000 labelled pairs by week 2.

Brief 18 — Viral RNA elements with 2026 structure tools

Difficulty 3 · Team 3–4 · GPU light · MVP (week 8): an element catalogue for one virus family combining covariation support, 2D predictions and 3D consistency, calibrated on elements with experimental structures.

Question. Which conserved RNA structural elements in the untranslated regions of one virus family can be modeled consistently by current tools, where do alignment-based conservation analysis and 3D prediction disagree, and which elements present candidate small-molecule pockets?

Data.

- One family chosen in week 1 (candidates: Flaviviridae 3' UTRs, Coronaviridae 5' and 3' UTRs, Picornaviridae IRES elements, or a plant virus family); genomes from NCBI Virus, dereplicated at 95% identity.
- Experimental control structures from the PDB (for example flavivirus exoribonuclease-resistant RNAs, the SARS-CoV-2 frameshifting element, HCV IRES domains); Rfam families covering viral elements; published chemical-probing data where available.

Tools.

- Alignment and covariation: MAFFT, LocARNA, Infernal (models built from Rfam seeds), R-scape, RNAalifold.
- Secondary structure: ViennaRNA, LinearFold, CONTRAfold.
- 3D: Boltz-2 (primary), Chai-1 (secondary); US-align and RNAdvisor.
- Pockets: SMARTPocket or Ribo-LENS if available, fpocket on RNA models as a baseline; PyMOL or ChimeraX for figures.

Metrics.

- Per element, reported before any prediction metric: alignment size, effective sequence diversity (Neff) and covariation support (R-scape E-values), so weak evidence is visible.
- Base-pair F1 between tool predictions and the alignment-derived structure.
- 3D consistency across seeds and tools (pairwise TM-score) and confidence; TM-score to experimental controls.
- Pocket analysis: detection rate on known ligand-binding elements (positive controls) and false-positive rate on elements without known ligands; any cavity in a predicted structure is reported as a hypothesis, never as evidence of binding; a confidence tier per element.

Week-by-week plan (item number = week).

1. Roles; choose the family; download and dereplicate genomes; collect control structures.
2. Alignment and element definition (Rfam models or LocARNA); R-scape covariation; alignment size and diversity recorded; refresh the closest prior work.
3. Freeze the protocol: element list, metrics, tiers, control analyses, tool versions.
4. 2D predictions per element across genotypes; agreement with the alignment-based structure.

5. 3D predictions with Boltz-2 (3 seeds) for each element representative and for the controls.
6. Consistency metrics and confidence analysis.
7. Comparison with experimental controls.
8. MVP (required): element catalogue for the family with evidence columns; internal review.
9. Main extension: second predictor; cross-tool agreement.
10. Main extension continued: pocket prediction calibrated on positive controls, then applied to the catalogue.
11. Stretch: genotype-level analysis, structural conservation against sequence variation.
12. Stretch: case studies of disagreements between alignment-based and 3D-based pictures.
13. Robustness: choice of representative, seeds, alignment method.
14. Freeze results; make figures.
15. Draft the report; package the catalogue.
16. Final report, repository and catalogue release, presentation.

Risks.

Risk	Fallback
Too many genomes	Dereplicate harder (90% identity) and sample per genotype.
Poor Rfam coverage for the family	LocARNA alignments plus R-scape; define elements de novo.
Weak covariation evidence for some elements	Shown in the evidence columns; such elements stay in the lowest tier.
Pocket tools unavailable, or a cavity read as binding	fpocket baseline only; pockets presented as hypotheses after calibration on positive controls.
Catalogue reads as descriptive only	Anchor every tier on the experimental controls and state what each tier licenses.

Scope.

Class	Work
Required (MVP)	One virus family: element catalogue with alignment size, diversity, covariation strength, 2D agreement and 3D consistency.
Main extension	A second predictor; pocket prediction calibrated on positive controls.
Stretch	Genotype-level analysis; case studies.

Novelty and reproducibility (pre-filled; complete by week 3)

Field	Starting point
Closest prior work	Rfam viral families; family-specific structure papers (flavivirus xrRNAs, the coronavirus frameshifting element, HCV IRES domains); SMARTPocket and Ribo-LENS (2026).
Specific new contribution	An evidence-graded catalogue combining covariation strength, 2D agreement and 3D consistency for one family, with pocket tools calibrated on positive controls.
Main endpoint	Number of elements per evidence tier, and 3D consistency (median pairwise TM-score) by tier.
Evaluation risks	Alignment quality, genome redundancy, overinterpreting pockets.
Feasibility evidence	Alignment and R-scape run on the family by week 2.

Field	Starting point

References (checked 16–17 October 2026)

Tool inputs and training cutoffs cited in the briefs, with the page each claim was read from.

- Boltz: input format documentation (custom alignment field marked protein-only; modifications field for protein, DNA and RNA chains); Boltz-2 model card quoting the report's training structures released before 1 June 2023.
- Chai-1: repository README (custom alignments as aligned.pqt files; no RNA-specific alignment route documented); training cutoff in the technical report, to be recorded in week 2.
- Protenix: README (RNA MSA supported from v1.0.0) and inference instructions (`--use_rna_msa`, search against NT-RNA, Rfam and RNAcentral).
- RNAPro: model page (alignment and template inputs).
- gRNAde: paper (16 designs per target, RhoFold self-consistency, designability thresholds).
- Rfam: rfam.org (release 15.1, January 2026, 4,227 families); Rfam 15 paper.
- RNA3DB: paper.

Benchmarks, assessments and datasets.

- CASP16; functional relevance of CASP16 nucleic acid predictions; CASP17 call for targets; RNA-Puzzles Round V.
- AlphaFold 3 for RNA benchmark; systematic benchmark of deep-learning RNA 3D methods (2024); limits of deep-learning RNA prediction methods (2026); FoldBench.
- RNAdvisor and RNAdvisor 2; RNArank; stereochemistry of RNA-Puzzles models.
- Szikszai et al., 2022; Szikszai et al., 2026; CHANRG; Ribonanza; Eterna100-V2.
- BEACON; zero-shot benchmark of 21 RNA language models; RNA-FM; RiNALMo.
- rMSA; elDORS; synthetic homologous sequences.
- Ribo-LENS; SMARTPocket; Runs N' Poses review; docking versus co-folding for proteins.
- RNA004 modification-calling benchmark; RMBase v3.0; m6A-Atlas; National Academies review.
- rnaglib structure–function benchmark; RNA–RNA interaction benchmark.